Alaska Landslide Sent a Tsunami 1,580 Feet Up a Mountain

From: ScienceDaily Most Popular Mined: 2026-10-20 5 min read

In August 2025, a mountainside in Alaska suddenly collapsed into a long, narrow inlet of the sea called a **fjord**. The falling rock created the second-highest **tsunami** in recorded history. Now scientists have described what happened and the warning signs that came before it.

The landslide happened in Tracy Arm, near the South Sawyer **Glacier**. Water and **debris** rushed 1,580 feet (481 meters) up the other side of the **fjord**. That is higher than the top floor of Taipei 101, a famous skyscraper in Taiwan. The force of the water scraped the **fjord**'s walls down to bare rock. Luckily, it happened just after 5 a.m., and no ships were nearby. A tour ship had visited the **glacier** only about 12 hours earlier.

Researchers at the Alaska Earthquake Center studied the event. They noticed a pattern: several big landslides in recent years have happened right where **glaciers** are shrinking. When the ice melts away, mountains seem to become less stable.

Before this landslide, there were clear clues. The **glacier** below had shrunk back more than a third of a mile in two months. Heavy rain had soaked into cracks in the mountain, which can push rock closer to breaking. And in the days before the collapse, instruments recorded thousands of tiny earthquakes, called **tremors**, coming from the slope.

The scientists believe these signs could have triggered alerts for nearby ships and buildings at least a day early, if a monitoring program had existed. They suggest a yellow-orange-red alert system, like those used for avalanches and volcanoes.

No landslide warning system of this size exists in the U.S. yet. The researchers say building one would take teamwork between state and federal **agencies**.

Word Treasure

fjord -- A long, narrow arm of the sea between steep cliffs, often carved by glaciers.

tsunami -- A series of very large waves caused when something suddenly moves a lot of water.

glacier -- A huge, slow-moving mass of ice that forms from packed snow over many years.

tremors -- Small shaking movements of the ground, like tiny earthquakes.

debris -- Broken pieces of rock, wood or other material left after something breaks apart.

agencies -- Government offices that are in charge of particular jobs, such as safety or science.

Background you need

A landslide happens when rock and soil on a steep slope suddenly break loose and slide downhill.

Glaciers can act like a support at the base of a mountain, so when the ice melts and shrinks back, the slope above can become less stable.

When a huge amount of rock crashes into water, it pushes the water away in powerful waves, which is how a landslide can cause a tsunami.

Instruments called seismometers record ground shaking, and scientists study those signals to spot unusual activity before and during disasters.

Break it down

LEAD: An Alaska landslide in August 2025 caused the second-highest tsunami on record
KEY EVENT: Rock fell into Tracy Arm fjord near South Sawyer Glacier
+- WHO: Researchers at the Alaska Earthquake Center
+- WHERE: Tracy Arm, Alaska
+- WHEN: Just after 5 a.m., about 12 hours after a tour ship visited
EVIDENCE: Glacier retreat, heavy rain and thousands of tiny tremors came before the collapse
PROPOSAL: A yellow-orange-red alert system like those used for avalanches and volcanoes
OPEN QUESTION: Can state and federal agencies build a landslide warning system in time for the next one?

Why it matters

As glaciers shrink, more mountains near the sea may become unstable, so learning to read warning signs could help protect ships, visitors and coastal towns.

Test what you remember

1. What caused the giant tsunami in Tracy Arm?

- (A) A meteor
- (B) An underwater volcano
- (C) A landslide falling into the fjord
- (D) A hurricane

2. How high did the water and debris rush up the fjord's other side?

- (A) 158 feet
- (B) 58 feet
- (C) 15,800 feet
- (D) 1,580 feet

3. Why was no one hurt by the wave, according to the article?

- (A) The wave was very small
- (B) It happened just after 5 a.m. and no ships were nearby
- (C) Everyone had been warned by sirens
- (D) A wall stopped the water

4. How far had the glacier below the slide retreated in two months?

- (A) About ten feet
- (B) It had grown instead
- (C) Five miles
- (D) More than a third of a mile

5. What did instruments record in the days before the collapse?

- (A) Rising air temperatures only
- (B) Loud thunder
- (C) Thousands of tiny earthquakes, or tremors
- (D) No activity at all

6. What do the scientists suggest building?

- (A) A new glacier
- (B) A taller sea wall in every town
- (C) A yellow-orange-red landslide alert system
- (D) More cruise ship routes

Answer key (try the quiz first!): 1: C 2: D 3: B 4: D 5: C 6: C

Questions to think about

1. Alaska Earthquake Center researchers: The warning signs could have triggered alerts at least a day early if a monitoring program had existed.

MY THOUGHTS (at least 20 words):